

Assignment 1

Unit-1

$$1. \text{ Sample Space } = \{ HHH, HHT, HTH, THH, TTT, TTH, THT, HTT \}$$

$$P(\text{Atleast 1 head}) = P(1 \text{ head}) + P(2 \text{ heads}) + P(3 \text{ heads})$$

= $\frac{7}{8}$ No. of total outcomes

$$= \boxed{\frac{7}{8}}$$

$$2. \text{ Sample Space } = 52 \text{ cards}$$

$$\text{No. of numbered cards} = 9 \times 4 = 36$$

$$P(\text{Numbered card}) = \frac{36}{52} = \boxed{\frac{9}{13}}$$

$$3. P(\text{Green}) = \frac{5}{12}, P(\text{Red}) = \frac{7}{12}$$

$$P(\text{Green first and Red second}) = \frac{5}{12} \times \frac{7}{11} = \boxed{\frac{35}{132}}$$

Q. 82

$$4. S = \left\{ \begin{array}{l} (1,1), (2,1), (3,1), (4,1), (5,1), (6,1) \\ (1,2), (2,2), (3,2), (4,2), (5,2), (6,2) \\ (1,3), (2,3), (3,3), (4,3), (5,3), (6,3) \\ (1,4), (2,4), (3,4), (4,4), (5,4), (6,4) \\ (1,5), (2,5), (3,5), (4,5), (5,5), (6,5) \\ (1,6), (2,6), (3,6), (4,6), (5,6), (6,6) \end{array} \right\}$$

$$P(\text{Sum of 7}) = \frac{6}{36} = \frac{1}{6}$$

5.

- (i) No. of honor cards in each suit = 4
Total no. of honor cards = 16

$$P(\text{Honor Cards}) = \frac{16}{52} = \frac{4}{13}$$

- (ii) No. of face cards per suit = 3
Total no. of face cards = $3 \times 4 = 12$

$$P(\text{face card}) = \frac{12}{52} = \frac{3}{13}$$

$$6. P(\text{Both are Diamond}) = P(A) = \frac{{}^{13}C_2}{{}^{52}C_2} = \frac{1}{17}$$

$$P(\text{Both are kings}) = P(B) = \frac{{}^4C_2}{{}^{52}C_2} = \frac{1}{221}$$

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$P(A \cap B) = \frac{7}{52} \cdot 0$$

$$P(A \cup B) = \frac{1}{17} + \frac{1}{221} - \frac{7}{52} \cdot 0 = \boxed{0.633}$$

7. $P(\text{Not getting a 4 in 1 die}) = \frac{5}{6}$

$$P(\text{Not getting a 4 in 3 dice}) = \left(\frac{5}{6}\right)^3 = \frac{125}{216}$$

$$P(\text{getting at least one 4 in 3 dice}) = \frac{91}{216}$$

8. $P(Q) = \frac{2}{7}$, $P(\bar{Q}) = \frac{5}{7}$

$$P(R) = \frac{4}{9}$$

$$P(\bar{R}) = \frac{5}{9}$$

$$P(\text{Not solving at all}) = \frac{5}{7} \times \frac{3}{7} \times \frac{5}{9} = \frac{25}{147}$$

$$P(\text{At least 3 solving}) = 1 - \frac{25}{147} = \boxed{\frac{122}{147}}$$

9. $P(X=2) = ?$, $p = \frac{1}{2}$, $q = \frac{1}{2}$, $n = 2$

$$P(X=r) = {}^nC_r \cdot p^r \cdot q^{n-r} = {}^5C_2 \times 0.5^2 \times 0.5^3 = \frac{5}{16} = \boxed{0.3125}$$

$$10. P(\text{Sum as } 22) = \left\{ (4,6,6,6), (6,4,6,6), (6,6,4,6), (6,6,6,4), (5,5,6,6), (6,5,5,6), (6,6,5,5), (5,6,6,5), (5,6,5,6), (6,5,6,5) \right\} = \frac{10}{1296}$$

$$P(\text{Sum as } 23) = \left\{ (5,6,6,6), (6,5,6,6), (6,6,5,6), (6,6,6,5) \right\} = \frac{4}{1296}$$

$$P(\text{Sum as } 24) = \frac{1}{1296}$$

$$P(\text{Sum is } 22+) = \frac{15}{1296} = \frac{5}{432} = \boxed{0.0115}$$

$$11. \text{Multiple of other} = \left\{ (1,1), (2,1), (3,1), (4,1), (5,1), (6,1), (1,2), (2,2), (4,2), (6,2), (1,3), (3,3), (6,3), (1,4), (2,4), (4,4), (1,5), (5,5), (1,6), (2,6), (3,6), (6,6) \right\}$$

$$P(\text{Multiple of other}) = \frac{22}{36} = \boxed{\frac{11}{18}}$$

$$12. P(\text{Each card from different suit}) = \frac{{}^4C_3 \times {}^{13}C_1 \times {}^{13}C_1 \times {}^{13}C_1}{{}^{52}C_3} = \frac{169}{4254} = \boxed{0.397}$$

13. No. of days in a leap year = 366
 No. of weeks in leap year = $\frac{366}{7} = 52.28$

So, there are 52 weeks and 2 days

To find the probability that there are exactly 52 Sundays we have to ensure of the 2 days, none are Sunday

$$S = \{(M, T), (T, W), (W, Th), (Th, F), (F, Sa), (Sa, Su), (Su, M)\}$$

$$\text{not Sunday} = \{(M, T), (T, W), (W, Th), (Th, F), (F, Sa)\}$$

$$P(\text{Exactly 52 Sundays}) = \frac{5}{7} = \boxed{0.714}$$

14. Circular permutation = $(15-1)! = 14!$

$$P(2 \text{ people sitting together}) = \frac{2! \times 13!}{14!} = \boxed{6:1}$$

15. Let Bags be A, B, C

$$P(A) = \frac{1}{3} = P(B), P(C)$$

$$P(R|A) = \frac{3}{10}, P(B|A) = \frac{7}{10}$$

$$P(R|B) = \frac{8}{10}, P(B|B) = \frac{2}{10}$$

$$P(R|C) = \frac{4}{10}, P(B|C) = \frac{6}{10}$$

$$P(C|R) = \frac{P(C) \cdot P(R|C)}{P(A) \cdot P(R|A) + P(B) \cdot P(R|B) + P(C) \cdot P(R|C)}$$

$$= \frac{\frac{1}{3} \times \frac{4}{10}}{\frac{1}{3} \times \frac{3}{10} + \frac{1}{3} \times \frac{8}{10} + \frac{1}{3} \times \frac{4}{10}}$$

$$= \frac{\frac{1}{3} \times \frac{4}{10}}{\frac{1}{3} \times \frac{3}{10} + \frac{1}{3} \times \frac{8}{10} + \frac{1}{3} \times \frac{4}{10}}$$

$$= \frac{4}{15}$$

16.

$$B_1 = 4W, 3B$$

$$B_2 = 3W, 5B$$

$$P(\text{Ball is Black from 2nd Bag}) = \frac{4}{7} \times \frac{5}{9} + \frac{3}{7} \times \frac{6}{9}$$

$$= \boxed{\frac{38}{63}}$$

17. $P(A) = 0.2$

$$P(B) = 0.16$$

$$P(C) = 0.14$$

$$P(A \cap B) = 0.08, \quad P(B \cap C) = 0.05$$

$$P(A \cap C) = 0.04$$

$$P(A \cap B \cap C) = 0.02$$

$$\begin{aligned} (i) \quad P(\bar{A} \cap \bar{B} \cap \bar{C}) &= 1 - P(A \cup B \cup C) \\ &= 1 - (0.2 + 0.16 + 0.14 - 0.08 - 0.04 + 0.02 - 0.05) \\ &= 0.65 \end{aligned}$$

$$\begin{aligned} (ii) \quad \text{Only } A &= P(A) - P(A \cap B) - P(A \cap C) + P(A \cap B \cap C) \\ &= 0.09 \end{aligned}$$

$$\begin{aligned} \text{Only } B &= 0.16 - 0.08 - 0.04 + 0.02 \\ &= 0.06 \end{aligned}$$

$$\begin{aligned} \text{Only } C &= 0.14 - 0.05 - 0.04 + 0.02 \\ &= 0.07 \end{aligned}$$

Total

$$\text{Total} = 0.09 + 0.06 + 0.07 = 0.22$$

$$= \boxed{22\%}$$